

HS 560: Time Series Econometrics

Course Code	HS 560
Course Name	Time Series Econometrics
L-T-P-C	2-0-2-3
Prerequisites	HS 550/Equivalent courses
Intended for	UG/PG Level
Mutual Exclusion	None
Approval	62nd BoA (Jan 2026)

Preamble

The main objective of this course is twofold: first, to make student academic and industry ready by enabling them to computing, programming and problem solving with basic programming skills where students get hands-on experience using R, an open-source language programming. Second, to engage students in active learning and developing their critical thinking about time series econometrics—based on economic theories using real time dataset, which is the need of the hour in the mainstream academia and industry. Thus, it helps students to acquaint with theory and application of time series econometrics. Importantly, the lab-based components will be conducted efficiently to the students with personal laptops.

Course Outcomes

At the completion of the course on time series econometrics the learner will be able to critically analyze the key concepts of time series and their application in the day-to-day decision making. They will be able to critically examine the nature and behavior of time series data and the forecasting ability—forecasting economic relationships, policy analysis. Moreover, it will help them understand how the theoretical concept could be put into an empirical one effectively.

Course Modules

- **Unit 1:** Data: Structure and Types || Data transformation—Trend, Seasonal, Cyclical and irregular/random components || Data transformation: natural and simple returns in financial time series || Time Series: basic concepts—stochastic processes, spurious regression, economic forecasting || Concept of stationary and nonstationary: Transforming non stationary time series—Difference stationary and trend stationary (8 Hours)
- **Unit 2:** Random walk model: with drift and without drift || Unit root tests: Dickey Fuller, Augmented Dickey Fuller (ADF), Phillips-Perron and KPSS unit root tests || AR(p), MA(q), ARMA(p, q), and ARIMA process. ARIMA modeling of time series data, Box-Jenkins approach Identification, estimation, diagnostic checking; Forecast function (12 Hours)
- **Unit 3:** Modeling Volatility – Understanding volatility and volatility clustering; Autoregressive Conditional Heteroscedasticity (ARCH) model; Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model || Cointegration approach: Engle-Granger cointegration test, Johansen cointegration test, Bayern and Hanck cointegration, Autoregressive distributed lag (ARDL) approach of cointegration || Error correction mechanism—Error correction term (10 Hours)
- **Unit 4:** Analyzing Multivariate Time Series Data Vector auto regression (VAR) model: Applications and problems; Impulse response functions, Variance decompositions, and Historical decomposition || Causality test: Granger causality and Toda-Yamamoto test || Vector error correction mechanism (VECM) (12 Hours)

Text Books

1. Chris Brooks (2008), *Introductory Econometrics for Finance*, 2nd Edition, Cambridge University Press.
2. Enders Walter (2014). *Applied Econometric Time Series*, 4th Edition. Wiley.

References

1. Zuur, A. F., Ieno, E. N., & Meesters, E. H. (2009). *A Beginner's Guide to R* (Vol. 722). New York: Springer.
2. Gary Koop, *Analysis of Economic Data*
3. Sankar Kumar Bhaumik (2015). *Principles of Econometrics. A modern approach using EViews*. Oxford.